



PAPER – 2

ADVANCED FINANCIAL MANAGEMENT



QUESTIONS

Risk Management

1. Mr. Y is a rational risk taker. He takes his position in derivative market of a single stock through margin trading for 4 days in a week. He does not take a position on Friday to avoid weekend effect and takes position only for four days in a week i.e. Monday to Thursday. He transfers the amount on Monday morning and withdraws the balance on Friday morning. He desires to take a maximum exposure in the derivative market where Value at Risk (VAR) should not exceed the balance lying in his bank account. The position by his manager, as per standing instructions, is taken on the free balance lying in the bank account in the morning on each Monday.

On Monday morning (before opening of the capital market) he has transferred an amount of ₹ 11 Crore to his bank account. A fixed deposit also matured on this Monday. The maturity amount of ₹ 63,42,560 was also credited to his account by the bank in the morning of the Monday. However, Mr. Y received the intimation of the same in the evening. The bank needs a minimum balance of ₹ 1,000 all the time.

The other information with respect to stocks A and B, which are under consideration for this week, is as under:

A		B	
Return (%)	Probability	Return (%)	Probability
6	0.10	4	0.10
7	0.25	6	0.20

8	0.30	8	0.40
9	0.25	10	0.20
10	0.10	12	0.10

From the information given above, choose the correct answer to the following questions:

- I. The amount that will not be considered for taking position on Monday morning is.....
 - (a) ₹ 63,41,560
 - (b) ₹ 63,42,560
 - (c) ₹ 11,00,00,000
 - (d) ₹ 10,99,99,000
- II. Which stock has a wider dispersion of returns based on the given probability distribution?
 - (a) Stock A
 - (b) Stock B
 - (c) Both have equal dispersion
 - (d) Cannot be determined
- III. In Value at Risk (VaR) analysis, which factor directly increases VaR for a given investment?
 - (a) Higher expected return
 - (b) Lower confidence level
 - (c) Higher standard deviation
 - (d) Shorter holding period
- IV. Since Mr. Y invests in derivative of a single stock (not a portfolio), the risk considered for VaR is.....
 - (a) diversifiable risk
 - (b) unsystematic risk only
 - (c) total risk

- (d) market risk only

Foreign Exchange Exposure and Risk Management

2. An Indian exporting firm, Radhey & Sons exported dry fruits worth of AUD 1 million to an importer in Sydney. Radhey and Sons are worried about likely depreciation of AUD in near future as it is likely that the export sum will be received after 3 months. Today as such as there is no derivative contract is available in AUD to hedge itself from such depreciation.

The following data is given:

Spot rate : ₹ 64.00/AUD

3 months interest rates (per annum):

	Deposit	Borrowing
₹	10%	12%
AUD	4%	5%

From the information given above, choose the correct answer to the following questions:

- I. In a money market hedge for a foreign currency receivable, the exporter should.....
 - (a) borrow in domestic currency and invest foreign currency
 - (b) borrow in foreign currency and invest domestic currency
 - (c) borrow and invest in domestic currency only
 - (d) borrow and invest in foreign currency only
- II. In case Radhey & Sons adopts Money Market hedge strategy, the interest rate to be used to compute amount borrowed shall be.....
 - (a) 4% per annum
 - (b) 5% per annum
 - (c) 10% per annum
 - (d) 12% per annum

- III. If a forward contract in AUD were available, the money market hedge would.....
- (a) always be superior
 - (b) always be inferior
 - (c) be compared with forward hedge for cost efficiency
 - (d) become invalid
- IV. Interest Rate Parity theory assumes.....
- (a) No capital mobility
 - (b) Presence of arbitrage opportunities
 - (c) Efficient markets with no arbitrage opportunities
 - (d) Fixed exchange rate system

Advanced Capital Budgeting Decisions

3. Tension Free Solutions (TFS) is a reputed is logistic service providers. All the deliveries are made by drivers of the mini trucks. The company employs many drivers, each of whom is provided with small trucks to make deliveries. Each driver drives approximately 40000 kms per annum. TFS decides to continue its present policy of always buying a new truck for its drivers but wonders whether the present policy of replacing the truck every three year is optimal or not. It is of believe that as new models are entering into market on yearly basis, it wishes to consider whether a replacement of either one year or two years would be better option than present three year period. The fleet of truck is due for replacement shortly in near future.

The purchase price of latest model truck is ₹ 5,50,000. Resale value of used truck at current prices in market is as follows:

Period	₹
1 Year old	3,50,000
2 Year old	2,10,000
3 Year old	90,000

Running and Maintenance expenses (excluding depreciation) are as follows:

Year	Road Taxes Insurance etc. (₹)	Petrol Repair Maintenance etc. (₹)
1	30,000	3,00,000
2	30,000	3,50,000
3	30,000	4,30,000

Using opportunity cost of capital as 10% you are required to determine optimal replacement period of truck.

Security Valuation

4. The data given below relates to a convertible bond:

Face value	₹ 1,000
Coupon rate	12%
No. of shares per bond	20
Market price of share	₹ 48
Straight value of bond	₹ 940
Market price of convertible bond	₹ 1,060

Calculate:

- (i) Stock value of bond.
 - (ii) The percentage of downside risk.
 - (iii) The conversion premium
 - (iv) The conversion parity price of the stock.
5. Mr. A is thinking of buying shares at ₹ 500 each having face value of ₹ 100. He is expecting a bonus at the ratio of 1: 5 during the fourth year. Annual expected dividend is 20% and the same rate is expected to be maintained on the expanded capital base. He intends to sell the shares at the end of seventh year at an expected price of ₹ 900 each. Incidental expenses for purchase and sale of shares are estimated to be 5% of the market price. He expects a minimum return of 12% per annum.

Should Mr. A buy the share? If so, advise the maximum price should he pay for each share?

Note: 1. Assume that there is no tax on dividend income and capital gains.

2. Present Value Factors should be used up to three decimal places, and the final calculations should be rounded off to two decimal places.

Business Valuation

6. The following data pertains to XYZ Inc. engaged in software consultancy business as on 31 December 2024.

(\$ Million)

Income from consultancy	935.00
EBIT	180.00
Less: Interest on Loan	<u>18.00</u>
EBT	162.00
Tax @ 35%	<u>56.70</u>
	<u>105.30</u>

Balance Sheet

(\$ Million)

Liabilities	Amount	Assets	Amount
Equity Stock (10 million share @ \$ 10 each)	100	Land and Building	200
Reserves & Surplus	325	Computers & Softwares	295
Loans	180	Current Assets:	
Current Liabilities	180	Debtors 150	
	<u> </u>	Bank 100	
	<u>785</u>	Cash <u>40</u>	<u>290</u>
			<u>785</u>

With the above information and following assumption you are required to compute

- (a) Economic Value Added (EVA)

(b) Market Value Added (MVA)

Assuming that:

- (i) WACC is 12%.
- (ii) The share of company currently is quoted at \$ 50 each.

Derivatives Analysis & Valuation

7. Following is the information available pertaining to shares of Omni Ltd.:

Current Market Price (₹)	₹ 840.00
Strike Price (₹)	₹ 900.00
Maximum Price (₹) expected in next 3 months' time	₹ 1,050.00
Minimum Price (₹) expected in next 3 months' time	₹ 756.00
Continuously Compounded Rate of Interest (p.a.) (%)	8.00%
e^{rt}	1.0202

Based on the above, calculate the value of the 3-month call option using the Binomial Method and the Risk-Neutral Method, and also draw a conclusion.

Note: Round off calculations upto 2 decimal points.

Interest Rate Risk Management

8. Suppose a dealer quotes 'All-in-cost' for a generic swap at 8% against six month SOFR flat. If the notional principal amount of swap is US\$ 5,00,000.

- (i) Calculate semi-annual fixed payment.
- (ii) Find the first floating rate payment for (i) above if the six month period from the effective date of swap to the settlement date comprises 181 days and that the corresponding SOFR was 6% on the effective date of swap.

In (ii) above, if the settlement is on 'Net' basis, how much the fixed rate payer would pay to the floating rate payer?

Note: 1. Generic swap is based on 30/360 days basis.

2. Round off the calculations to the nearest whole number.

Portfolio Management

9. The following information is available with respect of AK Ltd.

Year	AK Limited		Market		Return on Govt. Bonds
	Average Share Price (₹)	DPS (₹)	Average Index	Dividend Yield (%)	
2022	242	20	1812	4	6
2023	279	25	1950	5	5
2024	305	30	2258	6	4
2025	322	35	2220	7	5

Compute Beta Value of the company as at the end of 2025. What is your observation?

Note: Round off all calculations upto 2 decimal points.

International Financial Management

10. ICL an Indian MNC is executing a plant in Sri Lanka. It has raised ₹ 400 billion. Half of the amount will be required after six months' time. ICL is looking an opportunity to invest this amount on 1st April, 2020 for a period of six months. It is considering two underlying proposals:

Market	Japan	US
Nature of Investment	Index Fund (JPY)	Treasury Bills (USD)
Dividend (in billions)	25	-
Income from stock lending (in billions)	11.9276	-
Discount on initial investment at the end	2%	-
Interest	-	5 per cent per annum
Exchange Rate (1 st April, 2020)	JPY/INR 1.58	USD/INR 0.014
Exchange Rate (30 th September, 2020)	JPY/INR 1.57	USD/INR 0.013

You, as an Investment Manager, is required to suggest the best course of option.

Note: Present all figures in billions and round off the calculations to four decimal places.

Mergers, Acquisitions and Corporate Restructuring

11. Long Ltd., is planning to acquire Tall Ltd., with the following data available for both the companies:

	Long Ltd.	Tall Ltd.
Expected EPS	₹ 12	₹ 5
Expected DPS	₹ 10	₹ 3
No. of Shares	30,00,000	18,00,000
Current Market Price of Share	₹ 180	₹ 50

As per an estimate Tall Ltd., is expected to have steady growth of earnings and dividends to the tune of 6% per annum. However, under the new management the growth rate is likely to be enhanced to 8% per annum without additional investment.

You are required to:

- Calculate the net cost of acquisition by Long Ltd., if ₹ 60 is paid for each share of Tall Ltd.
- If the agreed exchange ratio is one share of Long Ltd., for every three shares of Tall Ltd., in lieu of the cash acquisition as per (i) above, what will be the net cost of acquisition?
- Calculate Gain from acquisition.

Mutual Funds

12. On 1st April, an open ended scheme of mutual fund had 400 lakh units outstanding with Net Assets Value (NAV) of ₹19. At the end of April, it issued 5 lakh units at an opening NAV plus 2% load, adjusted for dividend equalization. At the end of May, 4 Lakh units were repurchased at the opening NAV less 2% exit load adjusted for dividend equalization. At the end of June, 60% of its available income was distributed.

In respect of April-June quarter, the following additional information is available:

Particulars	₹ in Lakhs
Portfolio value appreciation	515.67
Income of April	31.960
Income of May	46.125
Income for June	58.470

You are required to calculate:

- (i) Income available for distribution;
- (ii) Issue price at the end of April;
- (iii) Repurchase price at the end of May; and
- (iv) Net Asset Value (NAV) as on 30th June.

Note: Round off all calculations upto 4 decimal points.

Security Analysis

13. Closing values of NSE Nifty from 8th to 21st day of the month of January of the year 2026 were as follows:

Days	Date	Day	Nifty
1	8	THU	25877
2	9	FRI	25683
3	10	SAT	No Trading
4	11	SUN	No Trading
5	12	MON	25790
6	13	TUE	25732
7	14	WED	25665
8	15	THU	No Trading
9	16	FRI	25694
10	17	SAT	No Trading
11	18	SUN	No Trading

12	19	MON	25585
13	20	TUE	25232
14	21	WED	25158

Give detailed analysis (on the basis of your calculations) of 21- day Exponential Moving Average (EMA) of Nifty during the above period.

The previous day exponential moving average of Nifty can be assumed as 26024.

Note: Round off calculations upto 3 decimal points.

Theoretical Questions

14. Explain the key decisions falling within the scope of financial strategy.
15. Explain risk associated with the use of Blockchain technology.

OR

Explain characteristics of Venture Capital Financing.